INCH-POUND

MIL-DTL-55302/31K
9 January 2004
SUPERSEDING
MIL-C-55302/31J
7 June 1993

DETAIL SPECIFICATION SHEET

CONNECTORS, PRINTED CIRCUIT SUBASSEMBLY AND ACCESSORIES: KEYING ACCESSORIES

This specification is approved for used by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-55302.

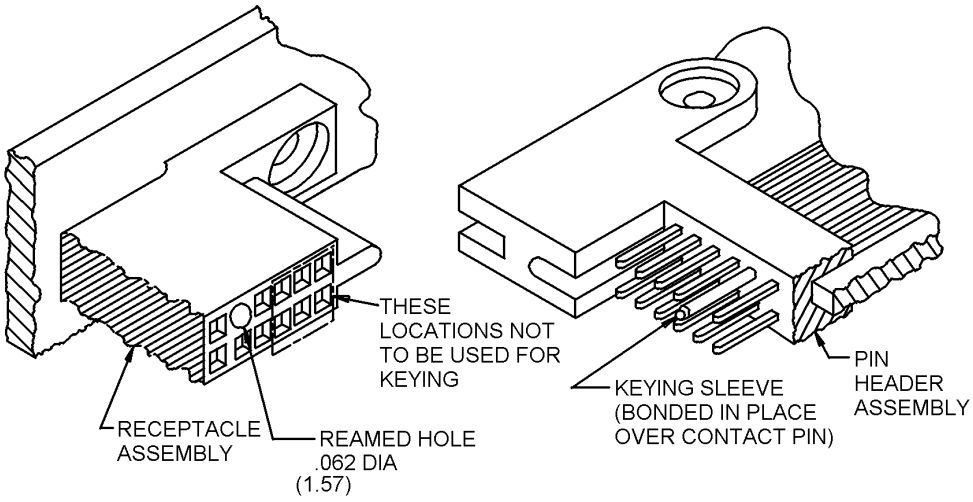


FIGURE 1. Installation of keying sleeve.

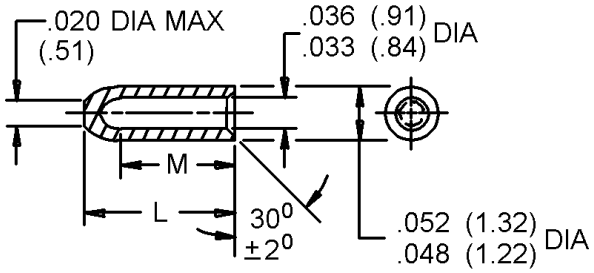


FIGURE 2. Keying sleeve.

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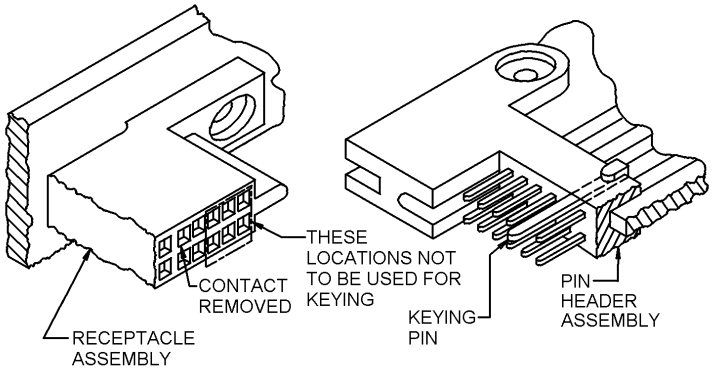


FIGURE 3. installation of keying pin.

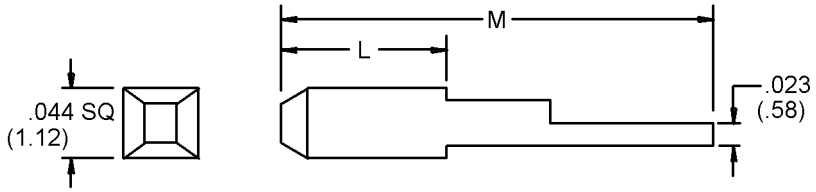


FIGURE 4. Keying pin.

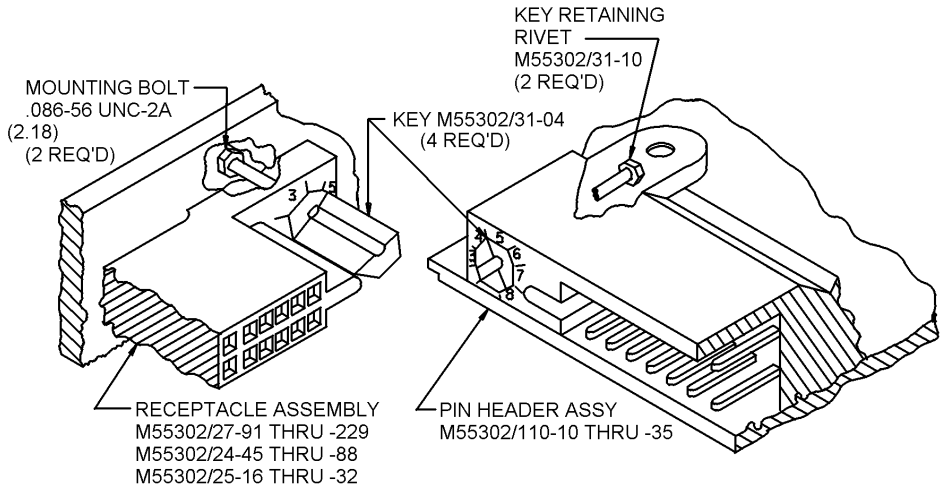


FIGURE 5. Installation of external key (position 6 shown).

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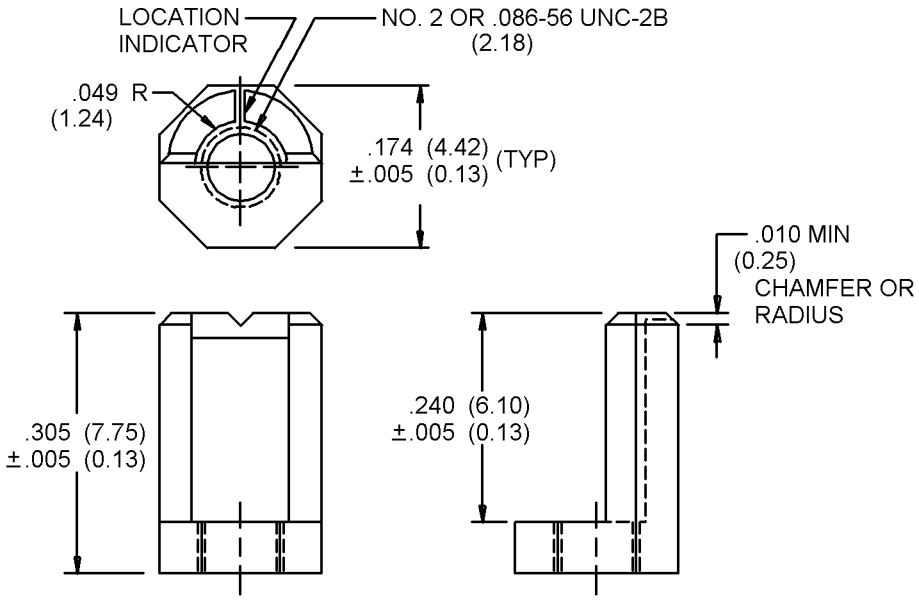


FIGURE 6. External key, M55302/31-04.

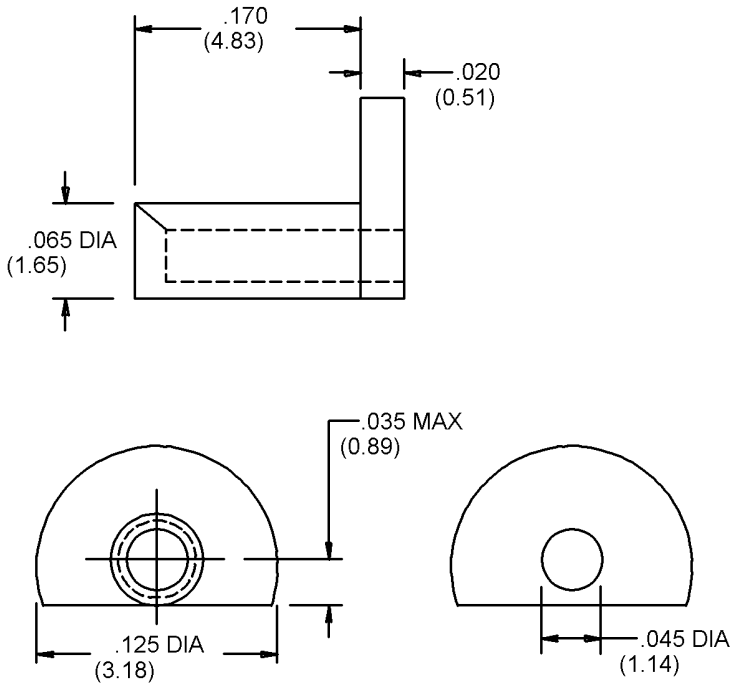


FIGURE 7. Retaining rivet for external key, M55302/31-05.

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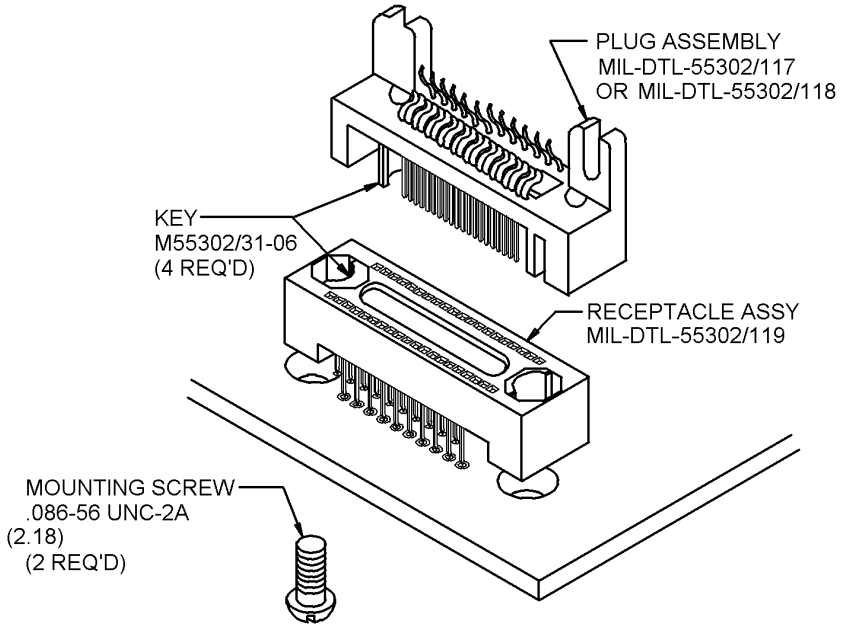


FIGURE 8. Installation of external key.

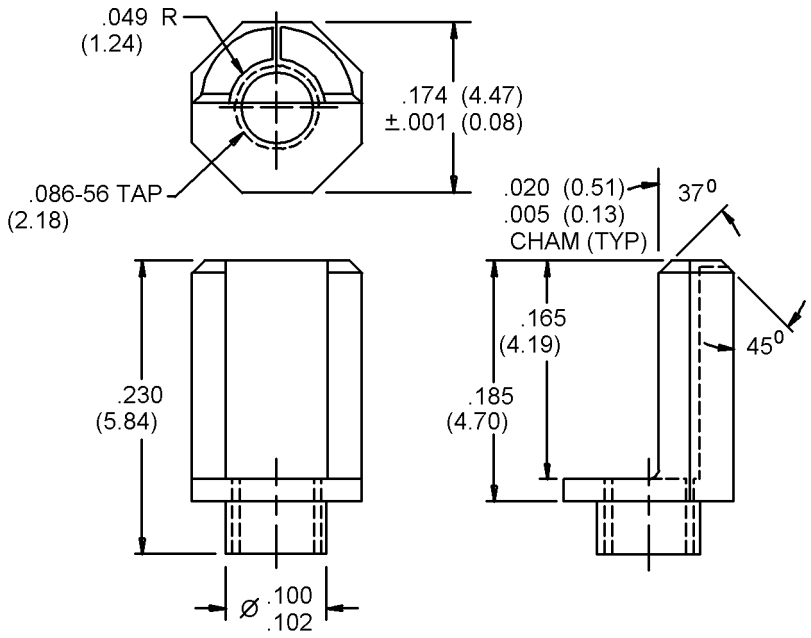


FIGURE 9. External key, M55302/31-06.

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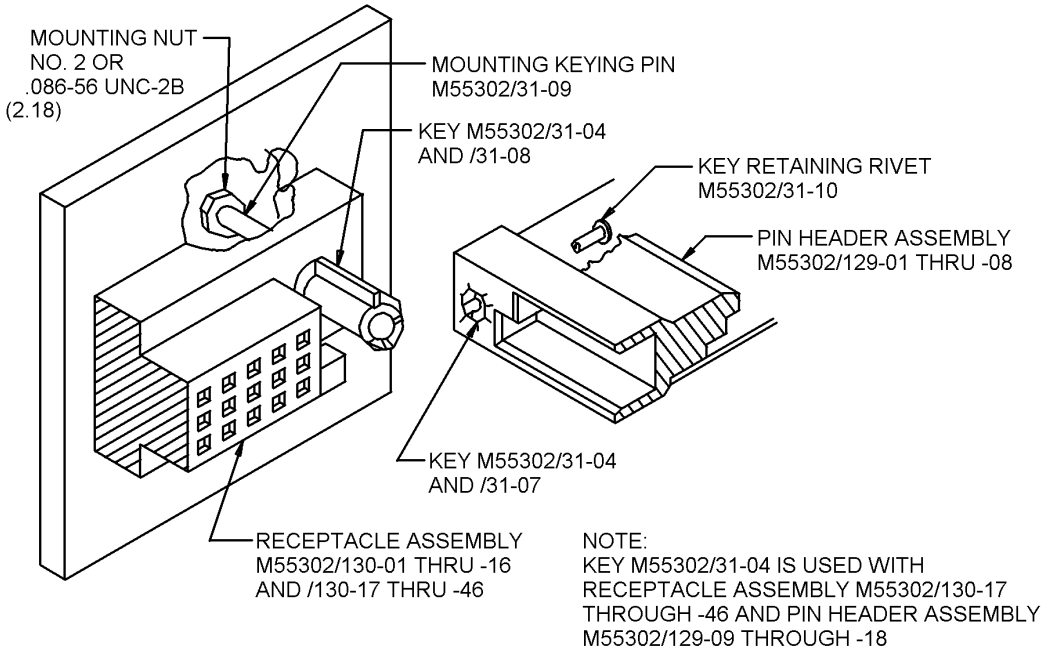


FIGURE 10. Installation of external key.

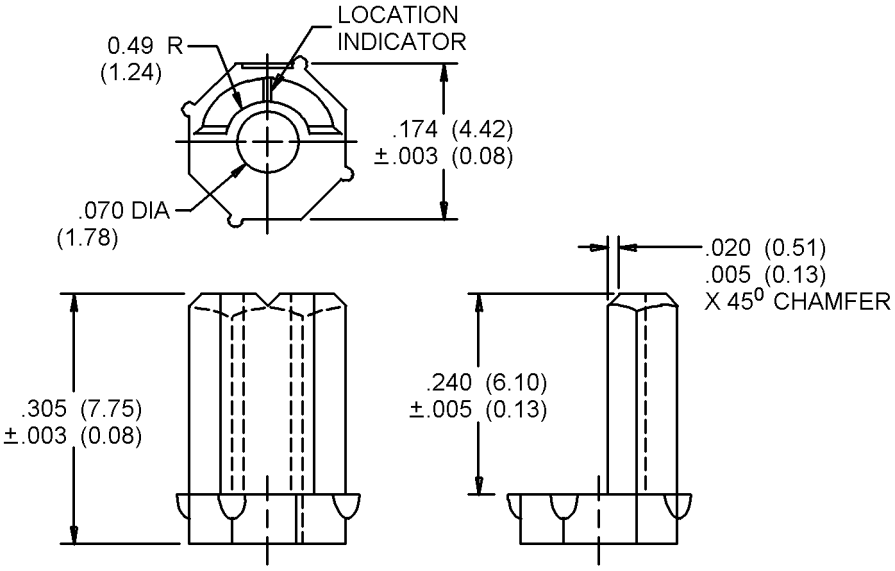


FIGURE 11. External key M55302/31-07.

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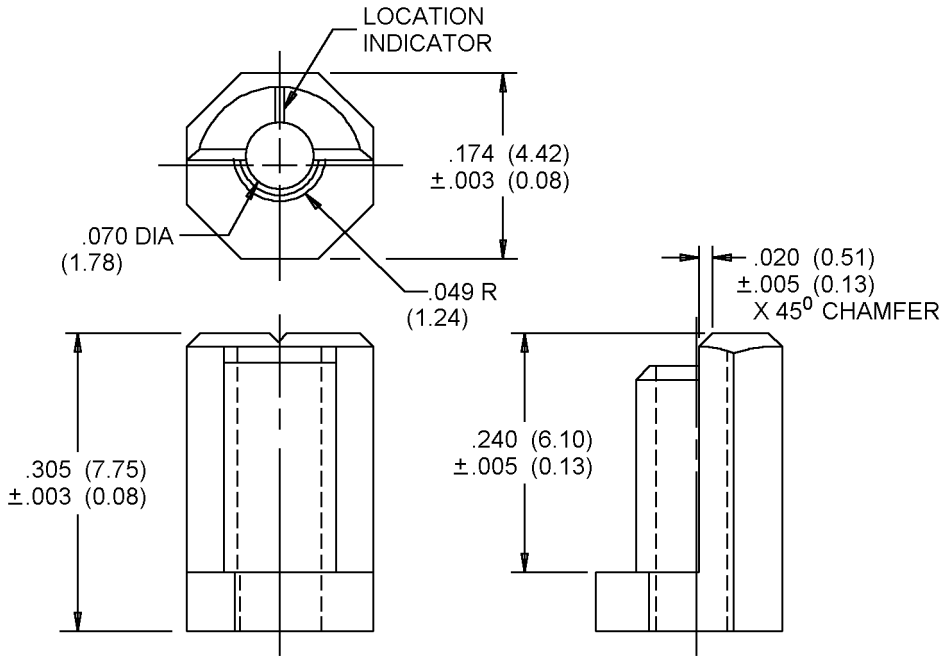


FIGURE 12. External key, M55302/31-08.

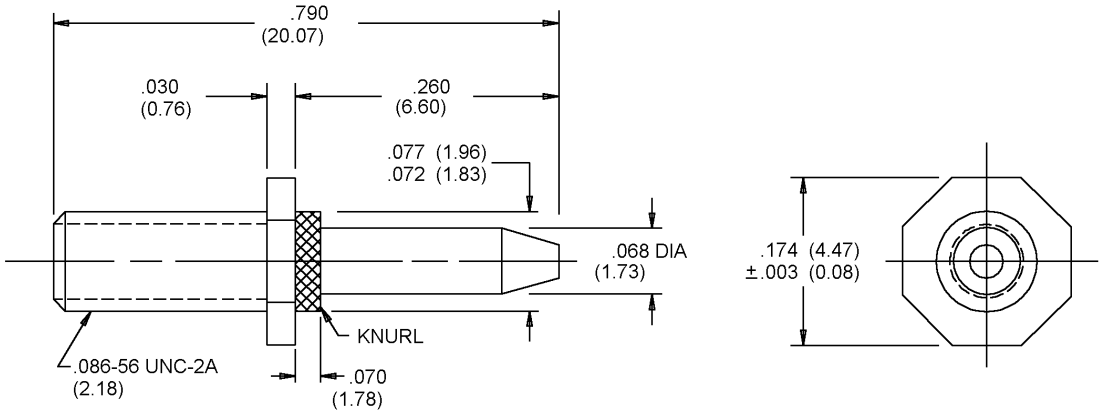


FIGURE 13. Mounting/keying pin, M55302/31-09.

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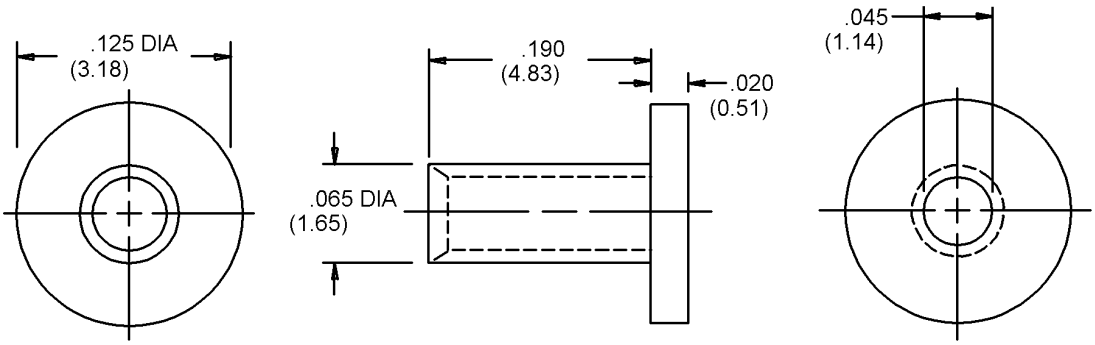


FIGURE 14. Retaining rivet for external key, M55302/31-10.

Notes and table for figures 1 through 14: See figure 15.

Type	Key PIN	L	M	Figure
I	M55302/31-01	.305 (7.75)	.240 min (6.10)	1, 2
I	M55302/31-02	.250 (6.35)	.255 min (5.72)	1, 2
II	M55302/31-03	.227 (5.77)	.700 ±.015 (17.78) ±(.38)	3, 4
III	M55302/31-04	---	---	5, 6
III	M55302/31-05	---	---	5, 7
III	M55302/31-06	---	---	8, 9
IV	M55302/31-07	---	---	10, 11
IV	M55302/31-08	---	---	10, 12
IV	M55302/31-09	---	---	10, 13
III	M55302/31-10	---	---	10, 14

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.005 (0.13 mm).
4. Metric equivalents are in parenthesis.

FIGURE 15. Notes and table for figures 1 through 14.

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REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figures 1 through 14.

Material:

Types I and II: Brass in accordance with Copper alloy C23000, C24000, C26000 and C26800 of ASTM B36 or C36000 of ASTM B16 or C35300 of ASTM B121 or C37700 of ASTM B124.

Type III:

Key: Stainless steel in accordance with SAE-AMS-QQ-S-763 or ASTM A582.

Rivet: Brass Copper alloy C23000, C24000, C26000 and C26800 of ASTM B36 or C36000 of ASTM B16 or C35300 of ASTM B121 or C37700 of ASTM B124.

Type IV:

Key: Glass filled polyester in accordance with ASTM D5927 type TPES013G30 or MIL-M-24519 type GPT-30F.

Mounting/keying pin: Stainless steel in accordance with SAE-AMS-QQ-S-763 or ASTM A582.

Finish:

Types I and II: Nickel in accordance with SAE-AMS-QQ-N-290.

Type III:

Key: Passivate in accordance SAE-AMS-QQ-P-35.

Rivet: Nickel in accordance with SAE-AMS-QQ-N-290.

Type IV

Mounting/keying pin: Passivate in accordance with SAE-AMS-QQ-P-35

Application:

Types I and II: This keying shall be used with connectors, as defined in MIL-DTL-55302/23, MIL-DTL-55302/24, MIL-DTL-55302/25, MIL-DTL-55302/26 and MIL-DTL-55302/27.

Type III: This keying shall be used with connectors, M55302/27-91 through M55302/27-229; M55302/110-10 through M55302/110-35, M55302/117-01 through M55302/117-11; M55302/118-01 through M55302/118-11; and M55302/119-01 through M55302/119-11, M55302/129-09 through M55302/129-18 and M55302/130-17 through M55302/130-46.

Type IV: This keying shall be used with M55302/129-01 through M55302/129-08 and M55302/130-01 through M55302/130-16.

Installation:

Type I: For receptacle assemblies MIL-DTL-55302/24, MIL-DTL-55302/25 and MIL-DTL-55302/27, remove contact and ream cavity to .062 inch diameter hole. For plug assemblies MIL-DTL-55302/23 or MIL-DTL-55302/26, bond keying sleeve with 3M adhesive (EC-2216 or equivalent) in place over contact (see figure 1).

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Type II: For receptacle assemblies MIL-DTL-55302/24, MIL-DTL-55302/25 and MIL-DTL-55302/27, remove contact. For plug assemblies MIL-DTL-55302/23 or MIL-DTL-55302/26, remove contact and insert keying pin as shown. Secure in place by bending tail over outside edge of housing. Trim excess material (see figure 3).

Recommended keying:

Types I and II: Two keys of same type located on opposite ends of connector excluding first six contacts on each end.

Types III and IV: (-04, -05 and -10): See figures 5 and 10.

Receptacle assembly: For Part or Identifying Numbers (PINs) M55302/27-91 through M55302/27-229, M55302/24-45 through M55302/24-88, M55302/25-16 through M55302/25-32 and M55302/130-17 through M55302/130-46, place key (-04) in numbered position as indicated by location slot (position 6 shown) and secure in place with .086-56 screw inserted through printed circuit board.

Plug assembly: For PINs M55302/129-09 through M55302/129-18, M55302/110-10 through M55302/110-35, M55302/23-16 through m55302/23-30 and M55302/26-10 through M55302/26-22, place key (-04) in the identically numbered position as indicated by location slot (position 6 shown) and secure in place with rivet M55302/31-05 or M55302/31-10, using AMP tool PIN 91117-1, 91117-3 or equivalent.

Type III (-06): See figure 8.

Receptacle assembly: For PINs M55302/119-01 through M55302/119-11, place key (-06) in one of the eight possible locations and press fit into mounting hole; then secure in place with .086-56 screw inserted through printed circuit board.

Plug assembly: For PINs M55302/117-01 through M55302/117-11, or M55302/118-01 through M55302/118-11, place key (-06) in the identical position as the receptacle (M55302/119-01 through M55302/119-08) and press fit into mounting hole.

Type IV (-07, -08, and -09): See figure 10.

Receptacle assembly: For PINs M55302/130-01 through M55302/130-16, mount connector to printed circuit board with mounting/key pin (-09) and secure with mounting nut. Place key (-08) over pin and press fit in to one of the eight possible locations.

Plug assembly: For PINs M55302/129-01 through M55302/129-08, place key (-07) in the identical position as the receptacle (M55302/130-01 through M55302/130-16) and secure in place with rivet M55302/31-10 using AMP tool PIN 91117-3 or equivalent.

PIN: M55302/31- (and dash number from table I).

Visual and mechanical inspections, required in accordance with MIL-DTL-55302.

Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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Referenced documents. In addition to MIL-DTL-55302, this document references the following:

MIL-DTL-55302/23	MIL-DTL-55302/31	MIL-DTL-55302/119	ASTM B16	SAE-AMS-QQ-N-290
MIL-DTL-55302/24	MIL-DTL-55302/110	MIL-DTL-55302/129	ASTM B36	SAE-AMS-QQ-P-35
MIL-DTL-55302/25	MIL-DTL-55302/117	MIL-DTL-55302/130	ASTM B121	SAE-AMS-QQ-S-763
MIL-DTL-55302/26	MIL-DTL-55302/118	MIL-M-24519	ASTM B124	
MIL-DTL-55302/27		ASTM A582	ASTM D5927	

CONCLUDING MATERIAL

Custodians:
Air Force - 11
DLA - CC

Review activities:
Air Force - 19, 99

Preparing activity:
DLA - CC

(Project 5935-4412-008)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at www.dodssp.daps.mil.

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